Q1 - 25 July - Shift 1

Let P be the plane containing the straight line $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$. If d is the distance of P from the point (2, –5, 11), then d^2 is equal to :

(A) $\frac{147}{2}$ (B) 96 (C) $\frac{32}{3}$ (D) 54

Space for your notes:

Q2 - 25 July - Shift 1

The line of shortest distance between the lines $\frac{x-2}{0}=\frac{y-1}{1}=\frac{z}{1}$ and $\frac{x-3}{2}=\frac{y-5}{2}=\frac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane P : ax – y – z = 0, (a > 0). If the image of the point (1, 1, –5) in the plane P is (α, β, γ), then α + β – γ is equal to ______. (Enter 0 if no such plane exists)

Space for your notes:

Q3 - 25 July - Shift 2

A plane E is perpendicular to the two planes 2x – 2y + z = 0 and x – y + 2z = 4, and passes through the point P(1, –1, 1). If the distance of the plane E from the point Q(a, a, 2) is $3\sqrt{2}$, then $(PQ)^2$ is equal to

(A) 9 (B) 12

(C) 21 (D) 33

Space for your notes:

Q4 - 25 July - Shift 2

The shortest distance between the lines $\frac{x+7}{-6}=\frac{y-6}{7}=z$ and $\frac{7-x}{2}=y-2=z-6$ is

(A) $2\sqrt{29}$
(B) 1
(C) $\sqrt{\frac{37}{29}}$
(D) $\frac{\sqrt{29}}{2}$

Space for your notes:

Q5 - 26 July - Shift 1

The length of the perpendicular from the point $(1, -2, 5)$ on the line passing through $(1, 2, 4)$ and parallel to the line $x + y - z = 0 = x - 2y + 3z - 5$ is :

(A) $\sqrt{\frac{21}{2}}$
(B) $\sqrt{\frac{9}{2}}$
(C) $\sqrt{\frac{73}{2}}$
(D) 1

Space for your notes:

Q6 - 26 July - Shift 1

Let Q and R be two points on the line $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point $P(4, 2, 7)$. Then the square of the area of the triangle PQR is______________.

Space for your notes:

Q7 - 26 July - Shift 2

A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j}, \hat{i}+\hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$ is

(A) $\frac{3\pi}{4}$
(B) $\frac{2\pi}{3}$
(C) $\frac{4\pi}{5}$
(D) $\frac{5\pi}{6}$

Space for your notes:

Q8 - 26 July - Shift 2

The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+a\hat{j}-\hat{k})$ and $\vec{r}=(\hat{i}+\hat{j})+\mu(-\hat{i}+\hat{j}-a\hat{k})$ from the point (2,1,4) is $\sqrt{3}$, is________.

Space for your notes:

Q9 - 26 July - Shift 2

The plane passing through the line L: $\ell x-y+3(1-\ell)z=1$, $x+2y-z=2$ and perpendicular to the plane $3x+2y+z=6$ is $3x-8y+7z=4$. If θ is the acute angle between the line L and the y-axis, then $415\cos^2\theta$ is equal to_____.

Space for your notes:

Q10 - 27 July - Shift 1

If the plane P passes through the intersection of two mutually perpendicular planes 2x + ky – 5z = 1 and 3kx – ky + z = 5, k < 3 and intercepts a unit length on positive x-axis, then the intercept made by the plane P on the y-axis is

(A) $\frac{1}{11}$ (B) $\frac{5}{11}$

(C) 6 (D) 7

Space for your notes:

Q11 - 27 July - Shift 1

Let the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}$ intersect the plane containing the lines $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ and $4ax-y+5z-7a=0=2x-5y-z-3$, $a\in\mathbb{R}$ at the point $P(\alpha,\beta,\gamma)$. Then the value of $\alpha+\beta+\gamma$ equals ________.

Space for your notes:

Q12 - 27 July - Shift 2

If the length of the perpendicular drawn from the point P(a, 4, 2), a > 0 on the line $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$ is $2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a+\sum_{i=1}^{3}\alpha_i$ is equal to :

(A) 7 (B) 8

(C) 12 (D) 14

Space for your notes:

Q13 - 27 July - Shift 2

If the line of intersection of the planes $ax+by=3$ and $ax+by+cz=0$, $a>0$ makes an angle $30°$ with the plane $y-z+2=0$, then the direction cosines of the line are :

(A) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$ (B) $\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0$

(C) $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$ (D) $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Space for your notes:

Q14 - 28 July - Shift 1

The foot of the perpendicular from a point on the circle $x^2+y^2=1$, $z=0$ to the plane $2x+3y+z=6$ lies on which one of the following curves ?

(A) $(6x+5y-12)^2+4(3x+7y-8)^2=1$, $z=6-2x-3y$

(B) $(5x+6y-12)^2+4(3x+5y-9)^2=1$, $z=6-2x-3y$

(C) $(6x+5y-14)^2+9(3x+5y-7)^2=1$, $z=6-2x-3y$

(D) $(5x+6y-14)^2+9(3x+7y-8)^2=1$, $z=6-2x-3y$

Space for your notes:

Q15 - 28 July - Shift 1

Let $P(-2,-1,1)$ and $Q\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagonal RS are $\alpha, -1, \beta$, where both α and β are integers of minimum absolute values, then $\alpha^2+\beta^2$ is equal to ________.

Space for your notes:

Q16 - 28 July - Shift 2

Let the lines $\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ be coplanar and P be the plane containing these two lines. Then which of the following points does **NOT** lies on P?

(A) $(0, -2, -2)$ (B) $(-5, 0, -1)$
(C) $(3, -1, 0)$ (D) $(0, 4, 5)$

Space for your notes:

Q17 - 28 July - Shift 2

A plane P is parallel to two lines whose direction ratios are –2, 1, –3, and –1, 2, –2 and it contains the point (2, 2, –2). Let P intersect the co-ordinate axes at the points A, B, C making the intercepts α, β, γ. If V is the volume of the tetrahedron OABC, where O is the origin and $p = \alpha+\beta+\gamma$, then the ordered pair (V, p) is equal to

(A) (48, –13) (B) (24, –13)
(C) (48, 11) (D) (24, –5)

Space for your notes:

Q18 - 29 July - Shift 1

If the foot of the perpendicular from the point A(– 1, 4, 3) on the plane P : 2x + my + nz = 4, is $\left(-2, \frac{7}{2}, \frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios 3, – 1, – 4, is equal to :

(A) 1 (B) $\sqrt{26}$
(C) $2\sqrt{2}$ (D) $\sqrt{14}$

Space for your notes:

Q19 - 29 July - Shift 1

Let a line with direction ratios a, –4a, –7 be perpendicular to the lines with direction ratios 3, –1, 2b and b, a, –2. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane $x - y + z = 0$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to ________.

Space for your notes:

Q20 - 29 July - Shift 2

Let Q be the foot of perpendicular drawn from the point P (1, 2, 3) to the plane $x + 2y + z = 14$. If R is a point on the plane such that $\angle PRQ = 60°$, then the area of ΔPQR is equal to:

(A) $\frac{\sqrt{3}}{2}$
(B) $\sqrt{3}$
(C) $2\sqrt{3}$
(D) 3

Space for your notes:

Answer Key

Q1 (C)	**Q2** (0)	**Q3** (C)	**Q4** (A)
Q5 (A)	**Q6** (153)	**Q7** (A)	**Q8** (2)
Q9 (125)	**Q10** (D)	**Q11** (12)	**Q12** (B)
Q13 (B)	**Q14** (B)	**Q15** (450)	**Q16** (D)
Q17 (B)	**Q18** (B)	**Q19** (10)	**Q20** (B)

Q1 (C)

$$a(x-3)+b(y+4)+c(z-7)=0$$

$$P: \quad 9a-b-5c=0$$

$$-11a-b+5c=0$$

After solving DR's $\propto (1,-1,2)$

Equation of plane

$$x-y+2z=21$$

$$d=\frac{8}{\sqrt{6}}$$

$$d^2=\frac{32}{3}$$

Q2 (0)

DR's of line of shortest distance

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 1 \\ 2 & 2 & 1 \end{vmatrix} = -\hat{i} + 2\hat{j} - 2\hat{k}$$

angle between line and plane is $\cos^{-1}\sqrt{\frac{2}{27}} = \alpha$

$\cos\alpha = \sqrt{\frac{2}{27}}$, $\sin\alpha = \frac{5}{3\sqrt{3}}$

DR's normal to plane $(1, -1, -1)$

$$\sin\alpha = \left|\frac{-a-2+2}{\sqrt{4+4+1}\sqrt{a^2+1+1}}\right| = \frac{5}{3\sqrt{3}}$$

$$\sqrt{3}\,|a| = 5\sqrt{a^2+2}$$

$$3a^2 = 25a^2 + 50$$

No value of (a) (No such plane exists)

Q3 (C)

(C)

Q4 (A)

(A)

Q5 (A)

d.r's of the line = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -2 & 3 \end{vmatrix} = \hat{i} - 4\hat{j} - 3\hat{k}$

$\therefore$ equation of line is

$\vec{r} = \hat{i} + 2\hat{j} + 4\hat{k} + \lambda\left(\hat{i} - 4\hat{j} - 3\hat{k}\right)$

Let A(1, 2, 4) and P be $(1+\lambda, 2-4\lambda, 4-3\lambda)$

$\therefore \overrightarrow{PA}.\left(\hat{i} - 4\hat{j} - 3\hat{k}\right) = 0$

$\lambda = \frac{1}{2}$

$\Rightarrow P\left(\frac{1}{2}, 2, \frac{-5}{2}\right)$

$|AP| = \sqrt{\frac{21}{2}}$

Q6 (153)

Let $(2\lambda - 1, 3\lambda - 2, 2\lambda + 1)$ be any point on the line

$(2\lambda - 5)^2 + (3\lambda - 4)^2 + (2\lambda - 6)^2 = 26$

$\lambda = 1, 3$

Q (1, 1, 3) ; R (5, 7, 7) ; P (4, 2, 7)

Area of triangle PQR = ½ $\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right|$

$= \sqrt{153}$

Q7 (A)

$$\vec{n}_1 = \hat{i} \times (\hat{i} + \hat{j}) = \hat{k}$$

$$\vec{n}_2 = (\hat{i} + \hat{k}) \times (\hat{i} - \hat{j})$$

$$= \hat{i} + \hat{j} - \hat{k}$$

Line of intersection along $\vec{n}_1 \times \vec{n}_2$

$$= \hat{k} \times (\hat{i} + \hat{j} - \hat{k}) = -\hat{i} + \hat{j}$$

D.R of $\vec{a} = -\hat{i} + \hat{j}$

D.R of $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$

$$\vec{a} \cdot \vec{b} = -3 \text{ and } \left(\vec{a} \wedge \vec{b}\right) = \theta$$

$$\cos \theta = \frac{-3}{\sqrt{2} \times 3}$$

$$\theta = \frac{3\pi}{4}$$

Q8 (2)

$$\vec{r}=\left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$$

$$\vec{r}=\left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$$

D.R's of plane containing these lines is

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & a & -1\\ -1 & 1 & -a\end{vmatrix}=\hat{i}(1-a^2)-\hat{j}(-a-1)+\hat{k}(1+a)$$

$$\vec{n}=(1-a)\hat{i}+\hat{j}+\hat{k}$$

One point in plane : (1, 1, 0)

$\therefore$ equation of plane is

$$(1-a)(x-1)+(y-1)+(z-0)=0$$

$$(1-a)x+y+z+a-2=0$$

$$\therefore D=\frac{|(1-a)2+1+4+a-2|}{\sqrt{(1-a)^2+1+1}}$$

$$\Rightarrow |5-a|=\sqrt{3}.\sqrt{a^2-2a+3}$$

$$\Rightarrow a^2+2a-8=0$$

$$\Rightarrow a=2,-4$$

$\therefore$ largest value of a = 2

Q9 (125)

$\vec{n}_1 = \ell\hat{i} - \hat{j} + 3(1-\ell)\hat{k}$

$\vec{n}_2 = \hat{i} + 2\hat{j} - \hat{k}$

Direction ratio of line $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \ell & -1 & 3(1-\ell) \\ 1 & 2 & -1 \end{vmatrix}$

$= (6\ell - 5)\hat{i} + (3 - 2\ell)\hat{j} + (2\ell + 1)\hat{k}$

3x – 8y + 7z = 4 will contain the line $(6\ell - 5)\hat{i} + (3 - 2\ell)\hat{j} + (2\ell + 1)\hat{k}$

Normal of 3x – 8y + 7z = 4 will be perpendicular to the line

$= 3(6\ell - 5) + (3 - 2\ell)(-8) + 7(2\ell + 1) = 0$

$\Rightarrow \ell = \frac{2}{3}$

$\therefore$ direction ratio of line $\left(-1, \frac{5}{3}, \frac{7}{3}\right)$

Angle with y axis

$\cos\theta = \frac{5/3}{\sqrt{1 + \frac{25}{9} + \frac{49}{9}}}$

$\cos\theta = \frac{5}{\sqrt{83}}$

$\therefore 415\cos^2\theta = \frac{25}{83} \times 415 = 125$

Q10 (D)

Two given planes mutually perpendicular

2(3k) + k (-k) + (-5) 1 = 0

k = 1, 5

but k < 3 So k = 1

Plane passing through these planes is

$2x + y - 5z - 1 + \lambda (3x - y + z - 5) = 0$

$$\frac{x}{\frac{5\lambda+1}{2+3\lambda}} + \frac{y}{\frac{5\lambda+1}{1-\lambda}} + \frac{z}{\frac{5\lambda+1}{\lambda-5}} = 1$$

Given $\frac{5\lambda+1}{2+3\lambda} = 1 \Rightarrow \lambda = \frac{1}{2}$

So intercept on y - axis = $\frac{5\lambda+1}{1-\lambda} = 7$

Q11 (12)

Equation of plane

$4ax - y + 5z - 7a + \lambda$ (2x – 5y – z – 3)= 0

this satisfy (4, -1, 0)

$16a + 1 - 7a + \lambda(8 + 5 - 3) = 0$

$9a + 1 + 10\lambda = 0$ .....(1)

Normal vector of the plane A is $(4a + 2\lambda, -1 - 5\lambda, 5 - \lambda)$ vector along the line which contained the plane A is

i – 2j + k

$\therefore 4a + 2\lambda + 2 + 10\lambda + 5 - \lambda = 0$

$11\lambda + 4a + 7 = 0$ ......(2)

Solve (1) and (2) to get a = 1, $\lambda = -1$

Now equation of plane

$x + 2y + 3z - 2 = 0$

Let the point in the line $\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z-3}{-4} = t$

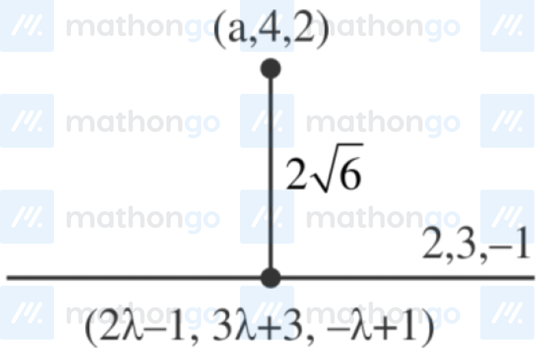
is $(7t + 3, -t + 2, -4t + 3)$ satisfy the equation of plane A

7t + 3 – 2t + 4 + 9 -12t – 2 = 0

t = 2

So $\alpha + \beta + \gamma = 2t + 8 = 12$

Q12 (B)



$(2\lambda - 1 - a)2 + (3\lambda - 1)3 + (-\lambda - 1)(-1) = 0$

$\Rightarrow 4\lambda - 2 - 2a + 9\lambda - 3 + \lambda + 1 = 0$

$\Rightarrow 14\lambda - 4 - 2a = 0$

$\Rightarrow 7\lambda - 2 - a = 0$

and,

$(2\lambda - 1 - a)^2 + (3\lambda - 1)^2 + (\lambda + 1)^2 = 24$

$\Rightarrow (5\lambda - 1)^2 + (3\lambda - 1)^2 + (\lambda + 1)^2 = 24$

$\Rightarrow 35\lambda^2 - 14\lambda - 21 = 0$

$\Rightarrow (\lambda - 1)(35\lambda + 21) = 0$

For, $\lambda = 1 \quad \Rightarrow a = 5$

Let $(\alpha_1, \alpha_2, \alpha_3)$ be reflection of point P

$\alpha_1 + 5 = 2 \qquad \alpha_2 + 4 = 12 \qquad \alpha_3 + 2 = 0$

$\alpha_1 = -3 \qquad \alpha_2 = 8 \qquad \alpha_3 = -2$

$a + \alpha_1 + \alpha_2 + \alpha_3 = 8$

Q13 (B)

$$\vec{n}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ a & b & 0\\ a & b & c\end{vmatrix}$$

$$= bc\hat{i}-ac\hat{j}$$

Direction ratios of line are **b, –a, 0**

Direction ratios of normal of the plane are **0, 1, – 1**

$$\cos 60^{\circ}=\left|\frac{-a}{\left|\sqrt{2}\right|\sqrt{b^2+a^2}}\right|=\frac{1}{2}$$

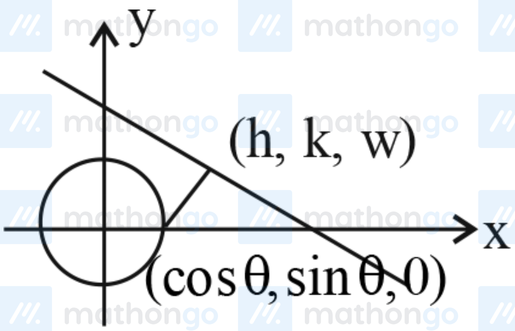
$$\Rightarrow \left|\frac{a}{\sqrt{a^2+b^2}}\right|=\frac{1}{\sqrt{2}}$$

$$\Rightarrow b=\pm a$$

So, D.R.'s can be $(\pm a, -a, 0)$

$\therefore$ D.C.'s can be $\pm\left(\frac{\pm 1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right)$

Q14 (B)



$$\frac{h-\cos\theta}{2}=\frac{k-\sin\theta}{3}=\frac{w-0}{1}$$

$$=\frac{-1(2\cos\theta+3\sin\theta-6)}{14}$$

$$h=\cos\frac{-2(2\cos\theta+3\sin\theta-6)}{14}$$

$$=\frac{10\cos\theta-6\sin\theta+12}{14}$$

$$k=\sin\theta-\frac{3}{14}(2\cos\theta+3\sin\theta-6)$$

$$k=\frac{5\sin\theta-6\cos\theta+18}{14}$$

Elementary $\sin\theta$ and $\cos\theta$

$$(5h+6k-12)^2+4(3h+5k-9)^2=1$$

Q15 (450)

$$RS \equiv (\alpha, -1, \beta)$$

$$\text{DR of } PQ \equiv \left(\frac{56}{17}+2, \frac{43}{17}+1, \frac{111}{17}-1\right)$$

$$\equiv \left(\frac{90}{17}, \frac{60}{17}, \frac{94}{17}\right)$$

$$\frac{90}{17}\alpha + \frac{60}{17}(-1) + \frac{94}{17}\beta = 0$$

$$90\alpha + 94\beta = 60$$

$$\beta = \frac{60-90\alpha}{94}$$

$$\beta = \frac{30(2-3\alpha)}{94}$$

$$\beta = -30\frac{(3\alpha-2)}{94}$$

$$\beta = \frac{-15}{47}(3\alpha-2)$$

$$\Rightarrow \frac{\beta}{-15} = \frac{3\alpha-2}{47}$$

$$\Rightarrow \beta = -15, \alpha = -15$$

$$\alpha^2 + \beta^2 = 225 + 225$$

$$= 450$$

Q16 (D)

Given, $L_1 : \frac{x-1}{\lambda} = \frac{y-2}{1} = \frac{z-3}{2}$

and $L_2 : \frac{x+26}{-2} = \frac{y+18}{3} = \frac{z+28}{\lambda}$

are coplanar

$$\Rightarrow \begin{vmatrix} 27 & 20 & 31 \\ \lambda & 1 & 2 \\ -2 & 3 & \lambda \end{vmatrix} = 0$$

$\Rightarrow \lambda = 3$

Now, normal of plane P, which contains L_1 and L_2

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ -2 & 3 & 3 \end{vmatrix}$$

$= -3\hat{i} - 13\hat{j} + 11\hat{k}$

$\Rightarrow$ Equation of required plane P :

$3x + 13y - 11z + 4 = 0$

(0, 4, 5) does not lie on plane P.

Q17 (B)

Normal of plane P :

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & -3 \\ -1 & 2 & -2 \end{vmatrix} = 4\hat{i} - \hat{j} - 3\hat{k}$$

Equation of plane P which passes through (2, 2,–2)

is $4x - y - 3z - 12 = 0$

Now, A (3, 0, 0), B (0, – 12, 0), C (0, 0, –4)

$$\Rightarrow \alpha = 3, \beta = -12, \gamma = -4$$

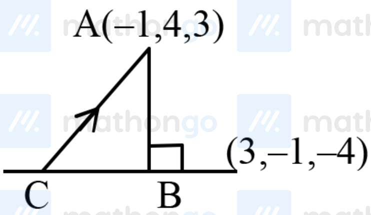
$$\Rightarrow p = \alpha + \beta + \gamma = -13$$

Now, volume of tetrahedron OABC

$$V = \left|\frac{1}{6}\overrightarrow{OA} \cdot \left(\overrightarrow{OB} \times \overrightarrow{OC}\right)\right| = 24$$

(V, p) = (24, –13)

Q18 (B)



Let B be foot of $\perp$ coordinates of $B=\left(-2,\frac{7}{2},\frac{3}{2}\right)$

Direction ratio of line AB is $<2,1,3>$ so

$m=1,\ n=3$

So equation of AC is $\frac{x+1}{3}=\frac{y-4}{-1}=\frac{z-3}{-4}=\lambda$

So point C is $(3\lambda-1,-\lambda+4,-4\lambda+3)$. But C lies

on the plane, so

$6\lambda-2-\lambda+4-12\lambda+9=4$

$\Rightarrow \lambda=1 \Rightarrow C(2,3,-1)$

$\Rightarrow AC=\sqrt{26}$

Q19 (10)

$(a,-4a,-7) \perp$ to $(3,-1,2b)$

$a = 2b$ $\quad\ldots$(i)

$(a,-4a,-7) \perp$ to $(b,a,-2)$

$3a + 4a - 14b = 0$

$ab - 4a^2 + 14 = 0$ $\quad\ldots$.(ii)

From Equations (i) and (ii)

$2b^2 - 16b^2 + 14 = 0$

$b^2 = 1$

$a^2 = 4b^2 = 4$

$$\frac{x+1}{5} = \frac{y-2}{3} = \frac{z}{1} = k$$

$\alpha = 5k - 1, \beta = 3k + 2, \gamma = k$

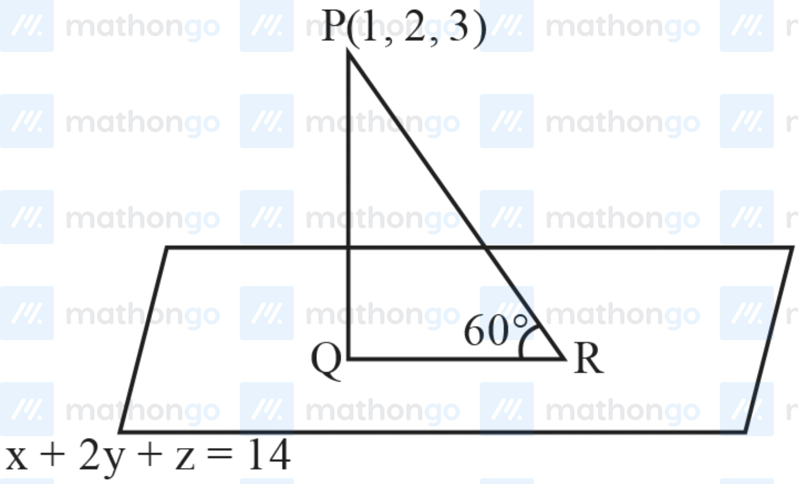
As (α, β, γ) satisfies $x - y + z = 0$

$5k - 1 - (3k + 2) + k = 0$

$k = 1$

$\therefore \ \alpha + \beta + \gamma = 9k + 1 = 10$

Q20 (B)



Length of perpendicular

$$PQ = \left|\frac{1+4+3-14}{\sqrt{6}}\right| = \sqrt{6}$$

$$QR = (PQ)\cot 60^\circ = \sqrt{2}$$

$$\therefore \text{ Area of } \Delta PQR = \frac{1}{2}(PQ)(QR) = \sqrt{3}$$